# Yifan Jiang

yifanjiang97@utexas.edu | Google Scholar Citation: 1165 | Website: yifanjiang19.github.io

## Education

**The University of Texas at Austin**, Austin, USA — 2020 – Present

*Ph.D.* in Electrical and Computer Engineering[1]
Advisor: Prof. Zhangyang (Atlas) Wang
Research Interests: Computational Photography, Generative Model, 3D Vision, Efficient ML

**Huazhong University of Science and Technology**, Wuhan, China — 2015 – 2019

*B.E.* in Electronic Information Engineering

## Publications

(* indicates equal contribution)

1. **Y. Jiang***, D. Xu*, P. Wang, Z. Fan, H. Shi, and Z. Wang. "SinNeRF: Training Neural Radiance Fields on Complex Scenes from a Single Image", European Conference on Computer Vision (**ECCV**) 2022.
2. **Y. Jiang***, Z. Fan*, P. Wang*, X. Gong, D. Xu, and Z. Wang. "Unified Implicit Neural Stylization", European Conference on Computer Vision (**ECCV**) 2022.
3. **Y. Jiang**, B. Wronski, B. Mildenhall, J. T. Barron, Z. Wang, and T. Xue. "Fast and High-Quality Image Denoising via Malleable Convolutions", European Conference on Computer Vision (**ECCV**) 2022.
4. **Y. Jiang**, X. Gong, Junru Wu, H. Shi, Z. Yan, and Z. Wang, "AutoX3D: Searching Ultra-Efficient Architecture for Video Understanding", Winter Conference on Applications of Computer Vision (**WACV**) 2022.
5. **Y. Jiang**, S. Chang, and Z. Wang, "TransGAN: Two Pure Transformers can Make One Strong GAN and That Can Scale Up", Advances in Neural Information Processing Systems (**NeurIPS**), 2021.
6. **Y. Jiang**, Z. He, J. Zhang, Y. Wang, Z. Lin, K. Sunkavalli, S. Chen, S. Amirghodsi, S. Kong, and Z. Wang, "SSH: A Self-supervised Framework for Image Harmonization", International Conference on Computer Vision (**ICCV**), 2021.
7. **Y. Jiang**, X. Gong, D. Liu, Y. Cheng, C. Fang, X. Shen, J. Yang, P. Zhou, and Z. Wang, "EnlightenGAN: Deep Light Enhancement without Paired Supervision", Transaction on Image Processing (**TIP**)
8. Z. Chen, **Y. Jiang**, D. Liu, and Z. Wang, "CERL: A Unified Optimization Framework for Light Enhancement with Realistic Noise", Transaction on Image Processing (**TIP**)
9. D. Xu*, P. Wang*, **Y. Jiang**, Z. Fan, and Z. Wang, "Signal Processing for Implicit Neural Representations", Advances in Neural Information Processing Systems (**NeurIPS**), 2022.
10. D. Xu, H. Poghosyan, S. Navasardyan, **Y. Jiang**, H. Shi, and Z. Wang, "ReCoRo: Region-Controllable Robust Light Enhancement by User-Specified Imprecise Masks", ACM Multimedia (**MM**), 2022
11. B. Pan, R. Panda, **Y. Jiang**, Z. Wang, R. Feris, and A. Oliva, "IA-RED2: Interpretability Aware Redundancy Reduction for Vision Transformers", Advances in Neural Information Processing Systems (**NeurIPS**), 2021.
12. Y. Fu, Z. Yu, Y. Zhang, **Y. Jiang**, C. Li, Y. Liang, M. Jiang, Z. Wang, and Y. Lin, "InstantNet: Automated Generation and Deployment of Instantaneously Switchable Precision Networks", Design Automation Conference (**DAC**), 2021.
13. T. Meng*, X. Chen*, **Y. Jiang**, and Z. Wang, "A Design Space Study for LISTA and Beyond", International Conference on Learning Representations (**ICLR**), 2021.
14. X. Gong, S. Chang, **Y. Jiang**, and Z. Wang. "AutoGAN: Neural Architecture Search for Generative Adversarial Networks", International Conference on Computer Vision (**ICCV**), 2019.

## In-submission

1. Q. Wu, X. Chen, **Y. Jiang**, and Z. Wang, "Chasing Better Deep Image Priors Between Over- and Under-parameterization", International Conference on Learning Representation (**ICLR**), 2023, In Submission.
2. Z. Fan, P. Wang, **Y. Jiang**, X. Gong, D. Xu, and Z. Wang, "NeRF-SOS: Any-View Self-supervised Object Segmentation from Complex Real-World Scenes", International Conference on Learning Representation (**ICLR**), 2023, In Submission.

[1] Studied at Texas A&M University from Aug. 2019 to Aug. 2020; then transferred with my advisor to UT Austin

3. **Y. Jiang**, P. Hedman, B. Mildenhall, D. Xu, J. T. Baron, Z. Wang, and T. Xue, "AligNeRF: Improved High-fidelity Neural Radiance Field via Alignment-aware Training", Computer Vision and Pattern Recognition (**CVPR**), 2023, In Submission.
4. **Y. Jiang**, Z. Xia, Z. Fan, C. Zhang, X. Zhang, and J. Chen, "Lidar-guided Video Depth Estimation on Smartphones", Computer Vision and Pattern Recognition (**CVPR**), 2023, In Submission.
5. D. Xu, **Y. Jiang**, W. Cong, P. Wang, Z. Fan, Y. Wang, and Z. Wang, "3D-Diffusion: Toward 3D Consistent View Generation via Diffusion Model", Computer Vision and Pattern Recognition (**CVPR**), 2023, In Submission.
6. Q. Wu, **Y. Jiang**, J. Wu, K. Wang, G. Zhang, H. Shi, Z. Wang, and S. Chang. "Grasping the Arrow of Time from the Singularity: Decoding Micromotion in Low-dimensional Latent Spaces from StyleGAN", Transaction on Image Processing, (**TIP**), In Submission.

## Internship Experience

**Adobe**, San Jose, USA — May. 2022 – Present

*Research Intern with Marc Levoy's Team, Adviser: Dr. Zhihao Xia, Dr. Cecilia Zhang, Dr. Jiawen Chen.*

- Working on monocular video depth estimation

**Google Research** , Mountain View, USA — May. 2021 – May. 2022

*Research Intern with GCam, Adviser: Dr. Tianfan Xue, Bart Wrongski, Dr. Ben Mildenhall, Dr. Jon Barron.*

- Developed a fast denoising network by predicting spatially-varying kernels at low resolution and using a fast fused op to jointly upsample and apply these kernels at full resolution. The resultant paper was accepted by ECCV'2022
- Designed a high-fidelity neural radiance field that can render high-qulity novel view images.

**Adobe**, San Jose, USA — May. 2020 – Nov. 2020

*Research Intern with Appled Research Team (ART), Adviser: Dr. He Zhang and Dr. Jianming Zhang.*

- Developed a self-supervised method for image harmonization that does not require human annotation labels. The resultant paper was accepted by ICCV'2021

**Bytedance AI Lab**, Beijing, China — Jan. 2019 – Aug. 2019

*Research Intern with US CV Lab, Adviser: Dr. Jianchao Yang and Dr. Xiaohui Shen and Dr. Ding Liu.*

- Designed a jointly image denoising and low-light enhancement algorithm, which was integrated in the selfie camera app FaceU.

## Media Highlight

- TransGAN was covered by Quanta Magazine (Mar. 2022) and was highlighted by Top AI influencers and high-profile YouTubers, as well as considered as the most influential new paper of the month (Feb. 2021).
- AutoGAN was covered by Synced AI Technology Industry Review (Aug. 2019), and also featured on Towards Data Science (Sep. 2019) and Analytics Magazine (Aug. 2019), etc.

## Community Services

- Reviewer for: CVPR'2021-2022, ICCV'2021, ECCV'2022, ICML'2022, NeurIPS'2022, ICLR'2023, Siggraph Aisa'2022, Siggraph'2022, WACV'2022, Transaction on Image Processing (TIP), International Journal of Computer Vision (IJCV), NeuroComputing, IEEE Robotics and Automation Letters (RA-L)
- Workshop Organizer for: ECCV RLQ-TOD Workshop 2020

## Invited Talks

- "Fast and High-Quality Image Denoising via Malleable Convolutions" at Adobe, Marc Levoy's team.
- "Vision Transformer for Image Generation, Editing, and Processing" at Google Research, Computational Photography Team.
- "TransGAN: Two Transformers Can Make One Strong GAN" at [cai-workshop], [SHI Lab @University of Oregon]